

From Proxemics Theory to Socially-Aware Navigation: A Survey

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Abstract In the context of a growing interest in modelling human behavior to increase the robots' social abilities, this article presents a survey related to socially-aware robot navigation. It presents a review from sociological concepts to social robotics and human-aware navigation. Social cues, signals and proxemics are discussed. Socially aware behavior in terms of navigation is tackled also. Finally, recent robotic experiments focusing on the way social conventions and robotics must be linked is presented.

Keywords Proxemics · Human-aware navigation · Socially-aware navigation

1 Introduction

The incursion of service and companion robots in our homes and workcenters opens a wide range of opportunities for mobile robotics applications. However, the ability of a robot to adapt its behavior according to social expectations will be determinant for the success of such applications. People and robots will navigate sharing the same physical space and rules must be identified to establish a social order. To be natu-

rally incorporated to human populated environments, mobile robots must be designed not only safe but sociable.

This article starts from the idea that people will keep the same conventions of social space management when they interact with robots than when they interact with humans. Researchers in social robotics that believe in that hypothesis can rely on the rich sociological literature to propose innovative models of social robots. A review of relevant concepts to human-aware navigation is presented in this article, starting from sociological notions and finishing with applications in the field of Social Robotics.

As outlined in Fig. 1, compared to a classical navigation framework made of navigation strategies in dynamic environments, a perception system and robot's constraints, a socially-aware robot navigation framework must manage a social layer. In the navigation block, the navigation algorithm receives an abstraction of the environment via the robot's sensors and/or external devices, in order to produce safe plans of navigation. The navigation solutions are returned to the robot which, using its dynamic model and controllers, sends commands to the actuators that modify its position in the environment. Social awareness is reached by the integration of both social conventions and a new set of techniques, dependent of the perception system, dealing with the processing of social behavior cues, semantics of space and prediction of behaviors. Social conventions are dependent on the particular environment, including culture (but this topic is not specifically addressed in this survey) but also on the robot physical properties and tasks.

In the present article, we consider that social conventions are behaviors created and accepted by the society that help humans to understand intentions of others and facilitate the communication. A focus is done on social conventions that are relevant to the robot navigation task, where robots must display not only safe but understandable behaviors.

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Socially-Aware Robot Navigation

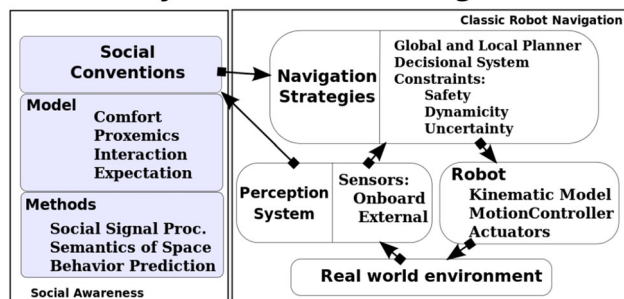


Fig. 1 The most important components of a socially-aware navigation system

The notion of safety addressed here is the physical one, i.e. avoiding collisions that could harm humans or the robot itself (as referred as safety-related stop space, safety-related object and safety-related obstacles in [44]). Navigation systems generally have a collision avoidance method. The point is that even if the robot has a very robust collision avoidance method, if the robot is not able to send social cues that permit humans to feel safe, the comfort of this latter will be affected.

The co-existence with robots will be more natural if mobile robots are able to recognize and respect social conventions.

To understand what humans expectations are, regarding to comfort-space relationship, implies to focus on the sociology field called *human spatial behavior*, which is the topic of Sects. 2 and 3. Section 4 presents a survey on the emergent field of social robotics, focused on socially-aware robot navigation, one of the main desirable characteristics of a social robot. The article finishes with conclusions in Sect. 5.

2 Social Behavior and Comfort

Human behavior has been studied from various perspectives: psychology, sociology, anthropology and neuroscience, each domain using different methodologies, scopes and evaluation criteria. In order to design a strategy for leading a robot safely in an environment populated by humans and at the same time preserving their comfort, it is necessary to explore how humans manage their surrounding space when they navigate and how their comfort could be affected by the motion of other pedestrians.

Hall [33] suggested that humans have modified the general proxemics rules followed by animals permitting a more complex social behavior. Humans do not attack nor escape when people invade their personal space. Social cues are there to notify others about the internal state of the affected individual. On the macroscopic level, territorial animals use scent marks to delimit their owned regions, humans have built walls, fences and borderlines. Designing robot behav-

iors according to animal behaviors has been addressed in the robotics community, for example in [6]. In the same sense [66] argued that the role of a social robot is close to the one of a domestic animal. Such view point is different from the one presented in this article. However, domestic animals have long shared their space with humans and ethological studies can provide interesting inputs for social robotics.

Automatic interpretation of human nonverbal communication is still very challenging. According to many authors like [42], the nonverbal communication represents more than sixty percent of the communication between two people or between one speaker and a group of listeners. Nonverbal communication is based on wordless cues that humans are sending and receiving constantly, mainly via visual modality. An emerging domain called Social Signal Processing (a survey can be found in [101]) aims to give the machines the ability to sense and understand that kind of social messages. In that framework social messages are defined in terms of social signals and social cues.

2.1 Social Cues for Social Signals

Definition 1 The term social cue describes a set of verbal or nonverbal messages. These cues, which can be facial expressions, body posture, proximity, neuromuscular and physiological activities, guide social interactions.

Definition 2 Human social signals are acts or structures that influence the behavior or internal state of other individuals. They are adaptive to the perceivers' response. They can transmit or not conceptual information or meaning [64].

Current technologies of sensing are already suited for detecting social cues. The main challenge is to link the social cues with the right social signal observed and to produce models for the social robotics field. Context plays a fundamental role for discovering the adequate association and will be addressed in Sect. 3.3.

Attention, empathy, politeness, flirting and agreement are all examples of social signals that can be detected analysing multiple social cues. Table 1 shows the major social cues and their relation with social signals.

This table shows for example that the social cue of hand gestures is related to the social signal of emotion. It is indeed possible to detect a person's stress by observing the quick repetitive motion of his/her fingers. Moreover, the importance of the vocal behavior cue is the way it is pronounced and not its meaning. However, regarding the visible social cues, one needs to be aware on the fact that the semantics of the body language varies with cultures.

The significative relation between social cues and social signals permits to link perceivable traits to subjective concepts. According to Table 1, posture as well as facial expressions and gaze behavior have an influence on all the social

Table 1 Major social cues and their relation to social signals

Social cues	Social signals						
	Emotion	Personality	Status	Dominance	Persuasion	Regulation	Rapport
Physical appearance							
Height			✓	✓			
Attractiveness		✓	✓	✓	✓		✓
Body shape		✓		✓			
Gesture and posture							
Hand gestures	✓				✓	✓	✓
Posture	✓	✓	✓	✓	✓	✓	✓
Walking		✓	✓	✓			
Face and eyes behaviour							
Facial expressions	✓	✓	✓	✓	✓	✓	✓
Gaze behaviour	✓	✓	✓	✓	✓	✓	✓
Focus of attention	✓	✓			✓	✓	✓
Vocal behaviour							
Prosody	✓	✓			✓		
Turn taking			✓	✓		✓	✓
Vocalizations	✓	✓		✓	✓	✓	✓
Silence							✓
Space and Environment							
Distance		✓	✓		✓		✓
Seating arrangement				✓			✓

Taken from [101]

signals described. As we will see in next section these cues can give useful evidence to assess the degree of concordance between robot navigation and social conventions.

2.2 Factors Influencing the Psychological Comfort

Many theories on psychological comfort explain the relation between distance, visual behaviors and comfort [1, 2, 30]. The relation between intrusion and discomfort is observed as linear, indicating that each increment of intrusion produces a comparable increment in discomfort [36]. [36] explains also that when threat (or potential threat) is high, distances tend to be larger, especially for females. In casual conversations, people respect space related to that activity and only participants can go inside this space without causing discomfort [48]. In [96] subjects rated intermediate distances (between four and eight feet) as most comfortable, preferable and appropriate for interaction situations. Preferred distances between people interacting are included in an optimal range and any deviations from this range result in discomfort. In that sense the theory presented by [5] proposed that an equilibrium of intimacy exists, involving components like level of nearness, glance, intimacy of topic and amount of smiling. If the disturbance is in the direction of too much intimacy, avoidance forces will predominate, and the subject will exhibit social cues aiming to communicate inner states.

It seems that this theory is maintained even in virtual environments [7, 8].

Since comfort in the above context is a subjective notion, no sensor can measure it directly. Studies explaining the way distance, posture and visual behavior affect comfort in humans can be used to develop useful models for social robotics. Specifically, Proxemics, which was proposed in the sociology literature will be presented in the next section.

3 Proxemics: Human Management of Space

Definition 3 Proxemics is the study of spatial distances individuals maintain in various social and interpersonal situations. These distances vary depending on environmental or cultural factors. The term was first proposed by Hall in [33] to describe the human management of space.

Hall observed the existence of certain unwritten rules that lead individuals to maintain distances from others, and lead others to respect this distance. Humans navigate maintaining spaces for themselves comparable to those they imagine the others would prefer, a notion that is supported by the Theory of Mind (an analysis can be read in [53]).

It has been proposed that the management of space done by a single person is different to the one done by a group of

people. According to [53], the term “we-space” is the result of the coordinated engagement of interactants and is different from the perspective commonly studied of an individual acting agent.

A socially-aware navigation strategy should manage four kinds of spaces: the ones related to a single person, to groups of people interacting, to human-object interaction and to human-robot interaction.

3.1 Spaces Related to Individuals

3.1.1 Personal Space

Definition 4 A personal space is the region around humans that they actively maintain into which others cannot intrude without causing discomfort [39].

A very complete historical review of the notion of personal space can be found in [89]. Figure 2 shows an environment with people that are not interacting explicitly. There is a natural arrangement of people motivated by the respect of individual personal spaces which are represented using circles (as proposed by Hall) although various shapes have been proposed in the literature, as illustrated in Fig. 3.

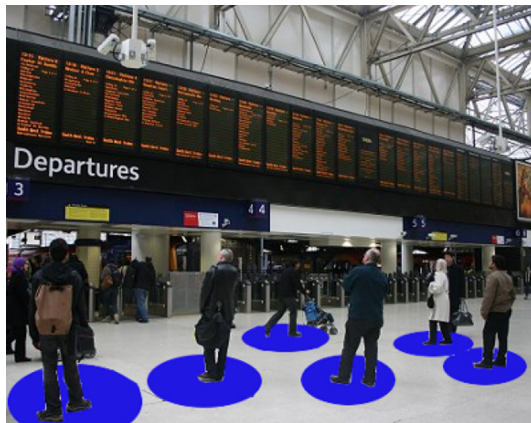
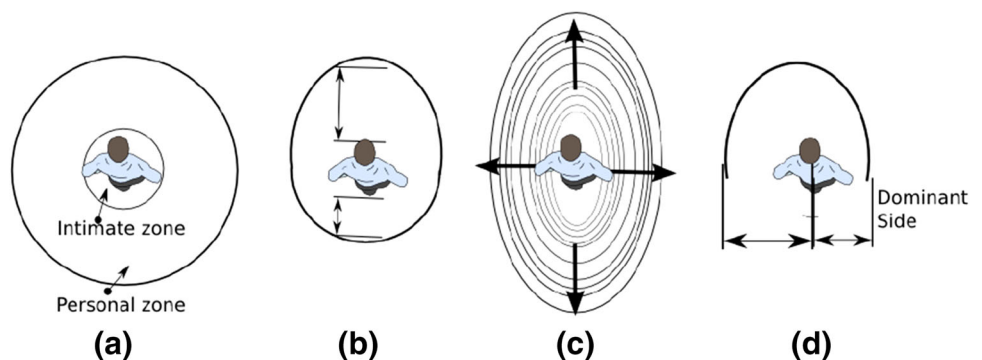


Fig. 2 Typical arrangement of humans observed as a consequence of respecting personal space (blue circles). (Color figure online)

Fig. 3 Different shapes of personal space: **a** Concentric circles [33]. **b** Egg shape, bigger in the front [37]. **c** Ellipse shape [40]. **d** Shape smaller in the dominant side [25]



Concentric Circles. According to [33] it is possible to classify the space around a person with respect to social interaction in four specific zones whose distances from human body are listed below:

- the public zone $> 3.6\text{m}$
- the social zone $> 1.2\text{m}$
- the personal zone $> 0.45\text{m}$
- the intimate zone $\leq 0.45\text{m}$

Obviously, these measures are not strict and vary with age, culture, type of relationship and context. As cultural differences affect the behavior related to spaces (some cultures avoid physical contact while others are more permissive), it is worthwhile to mention that the metrics proposed by Hall in his studies are valid for US citizens. Moreover, [9] describe a tolerance to intrusion for children which is related to their feelings of security with mothers or caregivers.

Egg Shape. People are more demanding regarding the respect of their frontal space, therefore frontal invasions are more uncomfortable [37].

Concentric Ellipses. Personal space refers to “the private sphere” in the Social Force Model proposed by [40]. The motion of pedestrians is influenced by other pedestrians by means of repulsive forces. The potential repulsive, according to this model, is a monotonic decreasing function with equipotential lines having the form of an ellipse that is directed into the direction of motion. The Social Force Model has been widely used to represent human behavior in agent simulation and has attracted the attention of the robotics community.

Asymmetric Shape. More recent work [25] claims that the size of the personal space does not vary according to the walking speed during circumvention of obstacles and that a personal space is asymmetrical, i.e., it is smaller in the pedestrian’s dominant side. They suggest that the personal space is used for navigating in cluttered environments. In that sense, [41] explored the fact that when someone wants to pass through an exiguous space, he evaluates the ratio between the size of the passage and the width of the body. That work supports the idea that there is a strong correlation

between the space representation that people have and their capabilities of actions. They divide the space around a human into two regions, the first which is in reach of the hand and a second one which is out of reach of the hand.

Other Related Aspects of Personal Space. Experiments presented in [38] support the idea that personal space is dynamic and situation dependent. It is considered that a personal space is a momentary spatial preference. Moreover, spatio-temporal models of personal space can be adjusted according to a velocity parameters [77]. A personal space is not only a psychological concept, recent work has provided some neuroscientific evidence that the amygdala may be implied in the regulation of interpersonal distances by triggering strong emotional reactions when for example, a personal space violation occurs [49].

More studies are needed to have a better idea of the three-dimensional shape of personal space and how it evolves over time. Quantitative models for shape, location and dynamics of personal space are interesting opportunities for collaborative research.

3.1.2 Information Process Space

Definition 5 An information Process Space (IPS) is the space within which all objects are considered as potential obstacles when a pedestrian is planning future trajectories [51].

In [51] the shape as well as the size of an information process space are explored. Authors explain that many current models of pedestrian movements share the information process space notion as common element. Experiments show that the information process space would have a cone shape instead of semicircular as proposed in similar works. Experiments consisted in collecting the gaze patterns of walking pedestrians. The results point out that the information process space is located in the exact front with a small relative lateral distance. Moreover the subjects in the study do not pay attention in the zone with an angle more than 45 degrees from the walking direction. Figure 4 shows a schema showing the information process space characteristics.

The information process space is strongly related to visual behavior. Work done by [5] shows that there is a strong relation between the eye-contact and the proximity, that is proximity grows with the eye-contact reduction. According to [29], approaching pedestrians can look at each others without embarrassment until their relative distance reaches approximately 2.5 meters, at this distance people typically look down.

However, it seems that the pedestrian's visual behavior is also conditioned by culture and gender as shown for example in the experimental study of [78] that compares behaviors of Japanese and American pedestrians. Experiments on

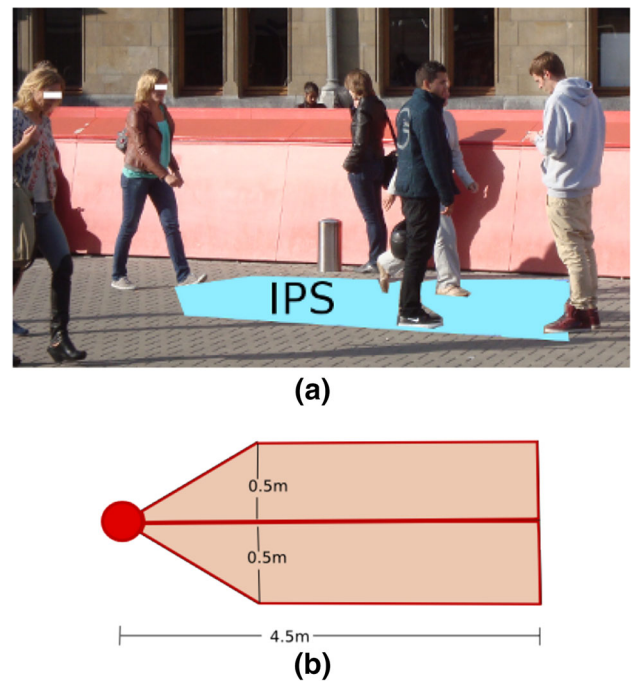


Fig. 4 The Information Process Space shape according to [51]. **a** A pedestrian is more interested in the exact front to detect obstacles and other pedestrians in order to calculate his next moves. **b** Shape and measures obtained for the IPS. The pedestrian is represented by the circle

obstacles avoidance in [41] suggest that vision gives information on space out of hand reach and controls locomotor action in a feed-forward manner. [100] proposed the concept of exosomatic visual architecture which allow agents guided by visual affordances to reproduce a natural movement well correlated with the observed human behavior.

3.2 Space Related to Groups of People

In [21, 52, 53], results converge to the idea that people keep more space around a group than the mere addition of single personal spaces. It is therefore important to study groups separately.

According to [29], humans react to a societal regulation through the concepts of focused and unfocused interactions.

Definition 6 Focused interaction occurs when individuals agree to sustain a single focus of cognitive and visual attention.

Definition 7 Unfocused interactions are interpersonal communications resulting solely by virtue of an individual being in another's presence.

Conversations are focused interactions because people share a common focus of attention with a shared common

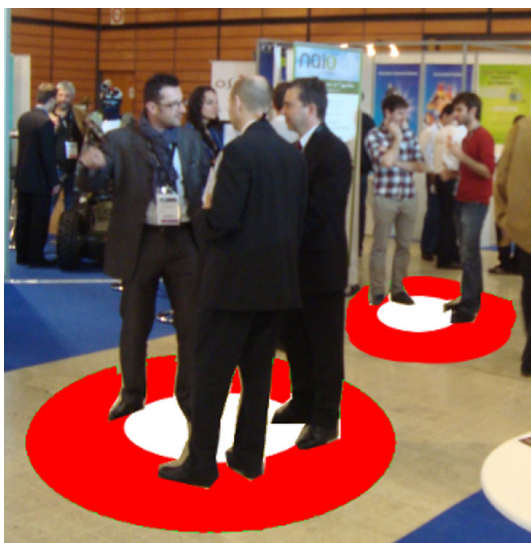


Fig. 5 Spatial patterns of arrangement formed by people interacting in groups. The O-Spaces and p-Spaces are represented respectively by white and red circles. (Color figure online)

space. In unfocused interactions people negotiate their position with others by means of nonverbal behaviors (like group arrangements) which improve the comfort and predictability of human actions.

The concepts of O-Space and F-formations [48] permit to detect conversations, both of them will be presented in the following section.

3.2.1 Interaction Spaces: The O-Space Concept

Definition 8 The O-Space is the joint or shared area dedicated to the main activity established by groups in focused interaction. Only participants can enter into it, they protect it and others tend to respect it, [48]. The O-Space geometrical characteristics depend on body size, posture, position and orientation of participants during the activity.

Definition 9 The p-Space is the space surrounding the O-Space which is used for the placement of the participant and their personal belongings [48].

Spatial patterns adopted by people in conversation act like social cues to inform pedestrians about the activity. Figure 5 illustrates how the position and orientation of people can help to decide what groups are in conversation and where would be located the O-Space. Social robots can take benefit from that knowledge to identify social interactions in the close environment.

Definition 10 The term F-formation is used to designate the system of spatial-orientation arrangement and postural behaviors maintained by people respecting their O-space [17].

The main functions of the F-formations are the regulation of social participation and the protection of the interaction against external circumstances. The shape of the F-formation varies according to the number of persons involved, their interpersonal relationship, the attentional focus and the environmental constraints (like furniture for example).

Most frequent F-formations concerning groups of two people have been identified by [17]. Table 2 gives a description of each one. [62] studies the support that physical spaces give to social interaction by using F-formations, observing that physical structures in the space can encourage and discourage particular kinds of interactions.

In general F-formations have been less studied than the personal space. A way to estimate the location of the O-space related to the F-formation by using position and head orientation is presented in [18]. Automatic collection of F-formation metrics are studied by [63]. Strategies to recognize formations and to integrate such knowledge in the robot's navigation decision process are proposed in [80] and [81].

3.2.2 Arrangements of Groups

Groups of Two People. According to [17, 48], two people conversing usually stand in one of the six following formations: N-shape, vis-a-vis, V-shape, L-shape, C-shape and side-by-side (see Table 2). The arrangements classification follows the criteria of body position and orientation, for example a Vis-a-vis formation is identified when both bodies face each, forming the letter H viewed from above. Table 2 analyses also the type of environment where the formation is more frequent. For example, N-Shape, Vis-a-vis and V-Shape are more frequent in open spaces not heavily used by pedestrians.

The vis-a-vis formation (also referred as face-to-face) is the basic mode of human sociality (see analysis done by [53]). Related to this, it has been shown that the perception of aperture between two people is constrained by psychosocial factors. For example, experiments in [68] showed that passing between a vis-a-vis formation is more comfortable if the passer knows well the people in formation.

Groups of More than Two People. When more than two people are in conversation, in absence of furniture, they commonly exhibit a circular shape arrangement (see Fig. 6). Therefore, in this case, the O-space is a circle whose center coincides with that of the inner space [48].

Efforts to provide an automated geometric method that detects social interactions by looking to interpersonal distance and torso orientation are presented in [32]. Social situations are modeled as the probability that at a certain place and time, there is a social situation among n persons given m social signals from these people.

Regarding the management of space done by a group of people interacting, [60] proposes that the interpersonal spa-

Table 2 Taxonomies of arrangements for a two-person formation defined by [17]

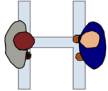
Environment	Arrangement	Description
Large and open spaces not heavily used by pedestrians		N-shape. Individuals face each other while they stand or sit maintaining their body planes parallel and slightly displaced by approximately half a body width
		Vis-a-vis. People face each other directly. When seen from above, it appears to form an array resembling a letter <i>H</i> with the body contours and noses
		V-shape. The participant's body planes intersect outside the formation at an angle of approximately 45°
Spaces that are semi-open and heavily trafficked by pedestrians		L-shape. Participants are standing at right angles to each other, with their body planes intersecting outside the gathering
Areas delineated by the presence of a large, solid and impenetrable object having little or no pedestrian movement		C-shape. Participants are standing at an angle of approximately 135° , and, when seen from above, they form the letter <i>C</i>
		Side-by-side. Two individuals face in the same direction but stand close enough to still have full access to each other's transactional segment

Fig. 6 Circular O-space in conversations for groups of more than two people

tial behavior is modulated not only by the distance between the interactants but also by the nature of the interaction (for example threatening or unthreatening social interactions).

It is important to mention the existence of a more general theory of spatial organization and classification proposed by [85] which structures the space based on the concept of human territoriality. The structured space influences and is sustained by a class of specific behaviors called territorial behaviors. In [24] groups traveling together can be discovered using a bottom-up hierarchical clustering approach that

compares sets of individuals based on proximity and velocity cues.

3.3 Spaces Related to Objects: The Activity and the Affordance Spaces

A human activity defines a virtual amount of space which is recognized and respected by the others. As a consequence, two more concepts must be incorporated in a socially-aware strategy: the activity and the affordance spaces.

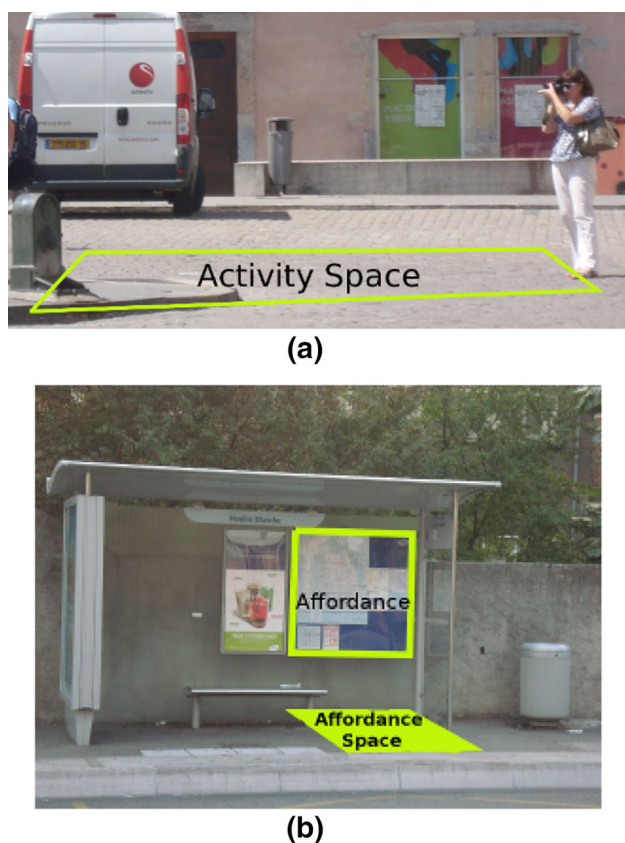


Fig. 7 Activity and affordance spaces. In **a** a woman is taking a picture, the space between her and her objective becomes an activity space. In **b** the bus schedule represents an affordance for humans and the space in front of this information becomes an affordance space

Definition 11 The Activity Space is a social space linked to actions performed by agents. The notion implies a geometric space but does not give an explicit definition for the shape. It can take multiple shapes depending on specific actions [59].

Definition 12 The Affordance Space is a social space related to a potential activity provided by the environment. In other words, Affordance Spaces are potential Activity Spaces.

An activity space is illustrated in Fig. 7a where a human is taking a picture. Normally, people in the surroundings bypass this space to avoid to interrupt the activity. An affordance space can generally be crossed without causing any disturbance (unlike an Activity Space) but blocking an affordance space could be socially not accepted. An example of affordance space is shown in Fig. 7b where the space in front of the bus schedule can be potentially used to read the information.

Related to the previous concepts, [13] claims that perceived geometrical features of the environment must be linked with semantic information of objects in order to achieve a semantic robot navigation. However the task becomes complicated as the perception of the environment done by sensors is objective (despite of uncertainty) while

human abstraction of the space is very subjective. It involves to infer, according to the context and the concepts defined above, what portion of empty space is restricted to robot navigation. In any case, it is necessary to take into account semantics of space in the planning of social acceptable navigation solutions.

Semantics of space and interaction with objects. Qualitative Spatial Reasoning is concerned with the acquisition, organization, utilization, and revision of spatial environments knowledge [12]. In the framework of space related to objects, [60] proposes the existence of a “practical space” created by the interaction of the space represented by the human body and the space where the objects exist. It can be translated into a practical rule: look around the human body to infer possible interactions with the world.

Other researchers propose similar ideas. For example, [31] explores the extension of the concept of proxemics between humans to proxemics with smart objects which can react to distance and orientation to adapt to humans. These ideas in form of proxemics applied to objects are presented also in [61] and in [103]. In the same sense, an ontology of spaces clustered by the ways humans interact with them is presented in [26]. Such approach is used to implement spatial cognition for robot-assisted shopping.

3.4 Robots and Proxemics

Research in robotics have lead to a set of social rules that will probably govern the robots’ physical behaviors when they interact with humans. It seems that the behavior of people sharing spaces with robots is not so different from the way they behave with other people [94]. A classification of the reviewed works is reported below according to the following proxemics factors: speed, appearance, direction of approach and other factors.

3.4.1 Speed

Experiments presented by [14] indicate that human subjects feel uncomfortable only with the robots’ fast approach speed. The comfortable speeds are between 0.254m/s and 0.381m/s while the uncomfortable fast speed is 1m/s. Normal walking speed for young human is about 1m/s, suggesting that humans prefer slower speed for a robot.

3.4.2 Appearance

According to [14] the appearance and size of the robot must be considered in the robot’s behaviors modeling. They observe humans approached and avoided by a robot, first by using the base robot only and then by adding a humanoid body to the base. Their results show that people who prefer the humanoid robot accept closer distances than the other

subjects. An empirical framework for Human-Robot proxemics is proposed in [102] where, after multiple experiments, a method is proposed to calculate the robot's approach distance estimate taking into account any combination of proxemics factors like robot appearance, human preferences or type of task. Such method consists in taking a base distance (57 cm) and calculating a distance of approach by adding the coefficient of proxemics factors which can be positive or negative.

Experiments done by [70] show that when a robot looks at people, these latter tend to increase their physical distance with the robot and this increase is even bigger when they dislike the robot. Moreover, men maintain a greater distance from the robot than women do.

3.4.3 Direction of Approach or Gaze Direction

According to [14], an indirect approach seemed to be preferred by experimented subjects. An indirect approach is a less threatening behavior because the threat of contact has been reduced.

In experiments done by [19], participants to which a robot has to bring an object, perceive the robot motion threatening and aggressive when the robot uses a direct frontal approach. Different conclusions are proposed by [99] where the user's evaluation shows that frontal approach directions (± 35 and 0 degrees with relation to the person orientation) are perceived as comfortable while farthestmost (± 70) directions are perceived as uncomfortable. Models for close, optimal and far distance to have a comfortable communication are extracted. In [43] they focus on the spatial interaction between a robot and a user analyzing the interaction using variations in distance and spatial orientation. They implement experiments based on the Wizard of Oz technique. Hall's interpersonal spatial zones and Kendon's formation are tested in human-robot interaction episodes in a home tour. In these experiments, users are asked to show a robot the location of objects and places. The Vis-a-Vis configuration is preferred to the other tested spatial configurations. In [56], they claim that it is possible to reconfigure an arrangement between a human and a robot by changing the position of the robot, when the robot is executing tasks of museum guide, which is more effective than only rotate its head.

[94] observe that the gaze direction has an effect on the minimum comfortable distance for people but the effect is different for women and men. When the robot's head is oriented towards the men's face, the distance needs to be smaller than when the robot's head is oriented towards the feet. On the contrary, for women, the distance needs to be higher.

Recently in [83] proactive gaze and automatic imitation are proposed as tools to quantitatively describe if and how human behaviors adapt in presence of robotic agents, based on the concept of motor resonance.

3.4.4 Other Factors

[94] show that people who have a personal experience with pets or robots need less personal space around robots than people who don't. Considering human-robot physical proximity, the taxonomy presented by [107] define six modes: none, avoiding, passing, following, approaching and touching. These modes are listed in increasing order of physical interaction degree.

What is not discussed in the present article is the fact that humans learn proxemic conventions along many years of social interaction, while robots will not have the same time to learn. Instead, datasets of human behavior along with dynamic and robust machine learning techniques can be used to give robots a minimal set of conventions.

There are still very few works focused on the human management of space around robots compared to the one around humans. However, robot proxemics is an increasing field and when successful cases of service robotics (like vacuum cleaners) become more numerous the new available scenarios will permit to corroborate or refuse the information presented in this section.

4 Social Robotics

In the robotic literature we can observe the growing interest in research topics including behavior of humans and its impact in robotic tasks. In this section we discuss the aspect of sociality from the point of view of the literature in robotics with a focus on mobile robots. As observed in the review, the keyword "social" in robotics and agents contexts is used in multiple and different ways. Consequently, it is complex to get a unique and complete definition of social robot. However, some important features emerge in the field of social robotics, as regards the human-robot interaction and socially-aware navigation.

4.1 Social Robot Abilities

[64] defines sociality as all the aspects that make individuals interact with each other to satisfy needs that could not be achieved by individuals alone. In contrast to the simple aggregation of individuals around favorable environmental conditions, sociality involves interactions between individuals. Social robots must engage in "natural" interaction with humans, i.e., interaction in the same way as humans do with other humans [20,86] and develop relationships or a rapport with them [46,47]. Robot may imitate human social norms and show a consistent set of behaviors [11] that have common sense [10]. Social robots must know how to initiate an interaction with a human [82], for example by displaying availability [106] or friendly attitude [35]. For [65] not

only natural initiation, but also maintenance, and termination of social interactions with humans are important. Moreover, robots exhibit their sociality by minimizing the interference with people in the same environment [84, 97]. Social robots must be proactive with the humans that are in their environment and behave as it is expected from them. The robot needs an internal understanding and adaptable social model of human society [23]. Social robots should be able to exhibit their status and intentions and to deal with their human partner's abilities and preferences [3].

Social skills applicable to social navigation are similar to those outlined above. The navigation of a social robot must consider the social aspects of interaction with people [74] and the comfort of humans, their preferences and their needs [88]. Social robot's behaviors must not afraid people and its motion intentions must be predictable (a.k.a. legibility) [54]. When social robots plan to navigate, they must be aware of the permitted and forbidden actions in social spaces and behave accordingly [59]. Their navigation involves an awareness of other users who are currently present or have been there in the past [45]. This implies that social robots must be able to distinguish obstacles from persons and behave in an appropriate way (for example, keeping comfortable distance from a person) [98]. Obviously, robots which are able to predict the behavior of the pedestrians can navigate in a more socially compliant way [55] and their movements will be easily understood and predicted. Therefore, people will trust and feel more comfortable with the robot [28]. Their safety will be enhanced also.

4.2 Socially-Aware Robot Navigation

Based on social robot notions and its abilities described above, the following definition can be proposed.

Definition 13 A socially-aware navigation is the strategy exhibited by a social robot which identifies and follows social conventions (in terms of management of space) in order to preserve a comfortable interaction with humans. The resulting behavior is predictable, adaptable and easily understood by humans.

Definition 13 implies, from the robot's point of view, that humans are no longer perceived only as dynamic obstacles but also as social entities.

Based on the key concepts proposed by [29] (see definitions 6 and 7) the related work on socially-aware navigation is divided into *focused interaction* and *unfocused interaction* regarding the main characteristics of each study (Table 3).

4.2.1 Unfocused Interaction

In unfocused interactions people and robot share the same environment and robots must negotiate their position with

Table 3 Related work on socially-aware robot navigation

Interaction	Related task and references
Unfocused interaction	Minimizing probability of encounter [16, 84, 97]
	Avoiding collisions [54, 57, 58, 72, 79, 95]
	Passing people [50, 55, 73, 75]
	Staying in line [71]
Focused interaction	Approaching humans [4, 15, 86, 105]
	Following people [28, 69, 108]
	Walking side-by-side [67]
Focused and unfocused interaction	Combination of previous listed tasks [34, 87, 93]

others by means of nonverbal behaviors or by the knowledge of rules in social spaces (Fig. 8).

Minimizing Probability of Encounter. In [97] a spatial affordance map is used to learn and predict spatio-temporal behavior of people in a house. Such map serves as a cost model for planning robot paths which minimize the probability of encounter with people. A very similar approach is presented in [84] where motion patterns are learned in an office environment by means of Sampled Hidden Markov Model. [16] propose a Spatial Behavior Cognition Model (SBCM), a framework to describe the spatial effects existing between humans and between a human and his environment. This SBCM is used to learn and predict (short-term and long-term) behaviors of pedestrians in an environment and to help a service robot to prevent potential collisions.

Avoiding Collisions. [95] propose a method for smooth collision avoidance of humans by using the social force model to determine whether a pedestrian intends to avoid a collision with the robot or not. In [72] an estimation of human's motion and personal space are used by a rescue robot to avoid collisions with evacuees. A recent work [79] extends the social force model by including a force due to face pose. The method is implemented in a robot which is able to avoid a human in a face-to-face confrontation. Based on their harmonious rules, [57] develop a Human-Centered Sensitive Navigation. Experiments show a robot avoiding humans and other robots by respecting its sensitive zones. In order to get legible strategies, experiments are conducted by [54] to collect data from human avoiding collision with other human. A focus is put on velocity adaptation more than on path adaptation. They propose a new cost model that takes into account the context in order to adjust the velocity of the robot.

A risk-based algorithm is employed in [90] for robot navigation in dynamic environments populated by human beings (Fig. 9), taking into account not only the risk of collision

Fig. 8 Examples of unfocused interaction. Robots must negotiate their position with others by means of nonverbal behaviors or by the knowledge of rules in social spaces

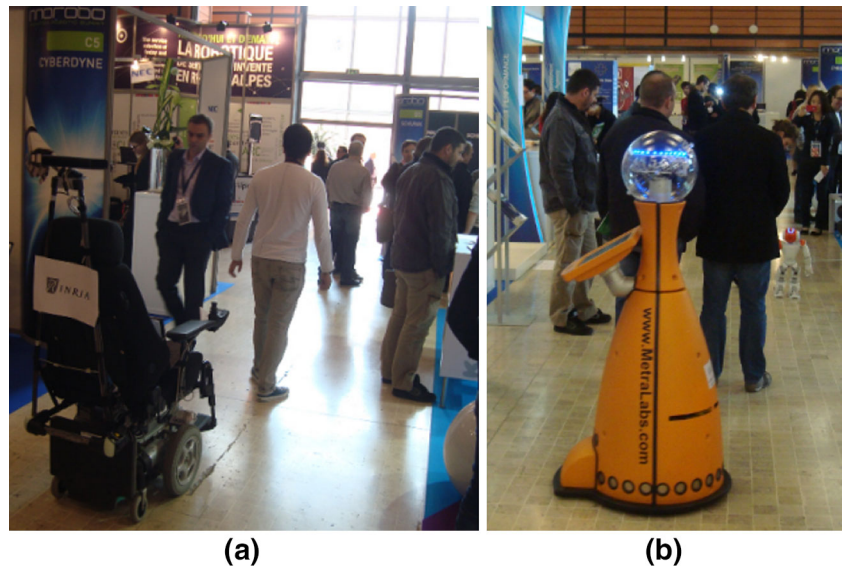
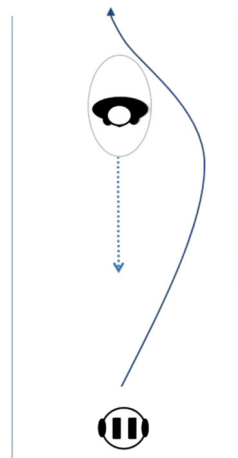


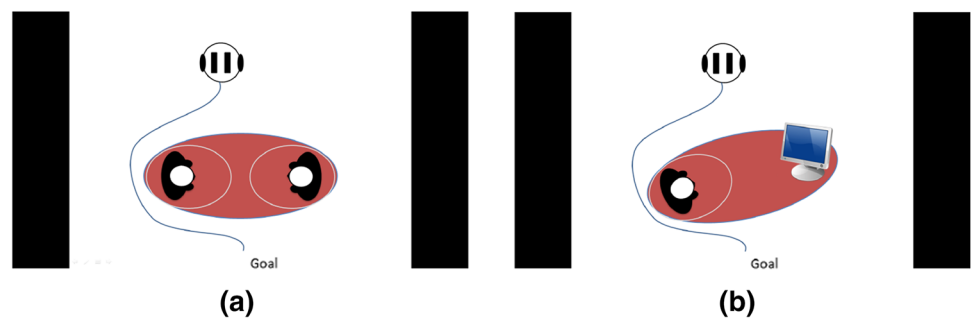
Fig. 9 A robot passing a person in a corridor taking into account the person's velocity



(already proposed in [22]) but also the risk of disturbance of human activities. The concepts of personal space, O-space and activity space (Fig. 10) are implemented in [80].

In [58], visual optimization of the path along with personal space are used to achieve human-like collision avoidance for agents in virtual crowds. Agents' speed is computed with the constraint of respecting a given minimal distance with other agents and obstacles.

Fig. 10 (left) A robot avoiding an interaction space. (right) A robot avoiding an activity space



Passing People. Inspired by human spatial behaviors, [73] presents a robot motion control which includes a module that achieves a people passing behavior in corridors (pass a person by the right). In [50] a generalized framework for representing social conventions as components of a constraint optimization problem is presented and used for path planning and navigation. Social conventions are modeled as costs to the A* planner with constraints like shortest distance, personal space and pass on the right. Simulation results show the robot navigating in a “social” manner, for example by moving to its right when encountering an oncoming person, as it is socially expected.

In [55] they propose a technique to reason about the joint trajectories that are likely to be followed by all the agents, including the robot itself. The approach learns a model of human navigation behavior that is based on the principle of maximum entropy from the observations of pedestrians. They implement their technique on a mobile robot and carry out experiments in which a human and a robot pass each other while moving to their target positions.

A socially aware mobile robot motion is addressed in [75]. The framework is supported by adding, deleting or modifying milestones based on static and dynamic parts of the environ-

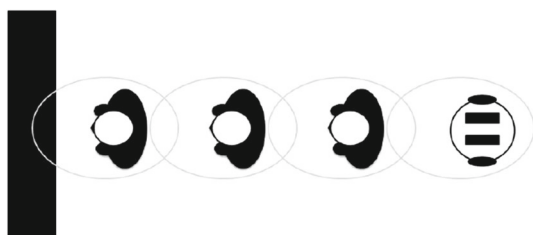


Fig. 11 A robot staying in line using a model of personal space model to determine its position

ment, the presence and the motion of an individual or group as well as various social conventions. Experiments show the robot that adapt dynamically its navigation around humans according the factors previously mentioned.

Staying in Line. Authors in [71] develop a model for the personal space of people standing in line in order to build a strategy for a robot to do the same task (Fig. 11). Their personal space model is used both to detect the end of a line and to determine how much space to leave between the robot and the person in front of it.

4.2.2 Focused Interaction

In focused interactions people and robot share a common focus of attention when executing their activity. In this kind of interaction the robot is expected to adapt naturally and dynamically to changes in the interaction (Fig. 12).

Approaching Humans. [15] investigate what abilities robots will need to successfully retrieve missing information from humans. A socially-aware navigation is employed to request help from human passers-by. An approach based on

Bezier curves is implemented as an nonlinear optimization problem with the objective to find a velocity profile for the Bezier path under constraints enhancing social acceptance. Experiments are conducted where the robot approaches a static human at different velocities and angles.

In [104, 105] formations are implemented to appropriately control the humanoid robot position as it presents information to a human. The model consists of the following constraints: proximity to a listener or to an object, listeners and presenter's fields of view.

In [4] the authors propose a method for a robot to join a group of people conversing. The results of the implementation and the experiments conducted with their platform show a human-like behavior (as judged by humans). The robot just wants to preserve the formation of the group and doesn't know explicitly where the o-space is located.

[86] study natural human interaction at the moment of initiating conversation in a shopkeeper scenario where a salesperson meets a customer. Then they use the observed spatial formation and participation state to model the behavior of initiating a conversation between a robot and a human.

Following People and Walking Side-by-Side. [28] compare two person following behaviors: one following the exact path of the person and other following in the direction of the person. They concluded that the second strategy is the most human-like behavior.

The people following behavior presented in [108] preserves socially acceptable distances from its human user (Fig. 13), and gives readable social cues (gaze, speech) indicating how the robot tries to maintain engagement during following.

[91] propose a partial planning algorithm to follow a leader. This leader is a human in the scene that, accord-

Fig. 12 Examples of focused interaction. People and robot share a common focus of attention



(a)

(b)

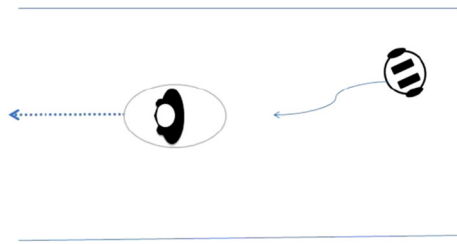


Fig. 13 The robot follows a person maintaining a socially acceptable distance

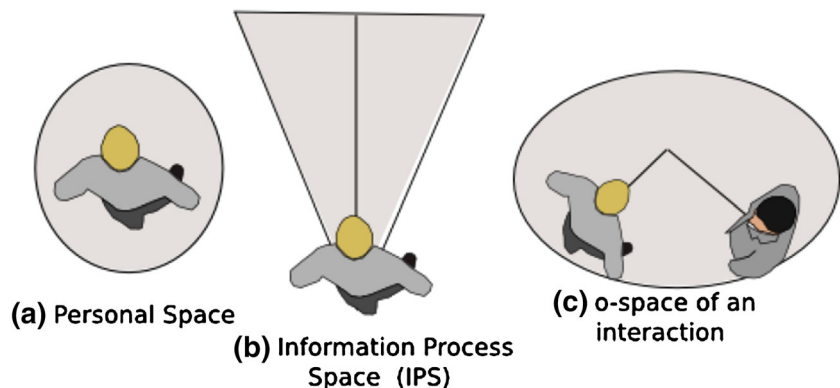
ing to a trajectory prediction method, goes toward the same location. The algorithm combines a person following and a mobile obstacle avoidance method. [69] propose an iterative planning technique that seeks for people moving to the same goal than the robot and follows them. The robot can get space to pass by shooing someone away in three steps: the robot approaches the person frontally, accelerates shortly and brakes again. In most cases, this behavior leads people to intuitively free the path. Even if this is not socially correct the robot is aware of human reactions to space invasions and makes use of them to navigate. In [67] they develop a computational model for side-by-side walking in a social robot by using an utility model describing how people prefer to move. The model was built based on recorded trajectories of pairs of people walking side-by-side.

4.2.3 Focused and Unfocused Interaction

Some works have proposed techniques capable of fulfill the two kinds of interaction.

Based on the person's pose and position, the adaptive system proposed in [34] detects if a person seeks an interaction with the robot or not. this work is presented as a basis for a human aware navigation. Navigation is implemented using human centered potential fields. This method is extended in [93] by including RRTs to minimize the invasion to social spaces of humans.

Fig. 14 We consider as discomfort the invasion made to humans' space by the robot, specifically, **a** Personal space **b** Information Process Space or **c** O-Space



[87] present a motion planner that takes into account explicitly the robot's human partners. The authors introduce criteria based both on the control of the distance between the robot and the human, and on the control of the robot's position within the human's field of view. The criterion of visibility proposed is based on the idea that the comfort increases when the robot is in the person's field of view.

5 Conclusion

This review is a guide, oriented to social robotics community, to begin projects related to socially-aware navigation. An introduction to important concepts related to social conventions are presented, first, from the point of view of sociology and after, from the point of view of robotics. Concerning the field of robot navigation, proxemics is the most investigated tool to improve the robot's sociality. However, as reviewed in Sect. 3, human management of space is a very complex dynamic system involving special factors for each one of the studied cases: one person, a group of people interacting or humans interacting with objects and robots. Context plays a paramount role for detecting social situations. For example, a social robot needs to identify an activity space from the data collected by sensors. A robot can decide to go through an activity space which is free of obstacles. Reflecting about the way a human would take navigation decisions in the same empty space, we observe that decision is taken not only based on safety (risk of future collision) but also by considering the meaning of that space associated to discomfort or disturbance to others. Social robots may take into account in their navigation schemes not only the personal space but also the information process space and the space of interaction (Fig. 14).

The link between proxemics and robotics literature need to be strengthened. Personal spaces are the most popular proxemics model used in robotics. Their shape are modeled according to works presented in [50,80], but literature in robotics proposes some extensions of these models

by subdividing this space into regions [75,92] to deal with human-robot interactions. Some roboticists have proposed new spaces to answer to robotics issues [88]. However, very few works in robotics tried to take into account proxemics factors such as speed, appearance, direction of approach, etc. to adapt dynamically the shape of social spaces [76]. A more complex model of space around humans is needed for social robot navigation, different from the standard technique which consists in inflating the obstacles detected from sensors by a predefined ratio. The reason is that perceived interpersonal distance is different from the physical one [27] and also because, as described in the sociological review, the inflation rate depends on a complex system regulating space through various communication modes.

The study of nonverbal behavior exhibited by humans can give cues for the robots to mimic the unfocused interaction resulting of the navigation close to humans and also to perform focused interaction in a more human-like manner. Perception of nonverbal behavior is, in general, a very challenging problem. Automatic techniques to collect social cues and methods to process them in order to get social signals are needed, not only for the robotics field but also for social sciences where it exists the requirement of both fairer judgment and more precise measurements.

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